

ERDŐS PROBLEM 928: QUANTITATIVE COMPARISONS AND ORIGINAL FORMULATIONS

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ABSTRACT. We show that OpenAI's joint Dickman law yields the density $\rho(1/\alpha)\rho(1/\beta)$ under the conditions stated in Erdős Problem 928. Here ρ is the Dickman–de Bruijn function. For fixed $0 < \alpha, \beta < 1$ and real $X \geq 2$, we compare the sets of integers specified by the problem and the theorem in the range $2 \leq n \leq X$. Their symmetric difference has size at most $\pi(X^\alpha) + \pi((X+1)^\beta)$, where π counts primes. This elementary bound handles both the strict inequalities and the shift of the second threshold from n^β to $(n+1)^\beta$. We also establish the density for Erdős's strict formulation in which both thresholds are powers of n , and compare the statements in Erdős's two 1976 papers and the 1978 Erdős–Pomerance paper. The analytic joint law is due to OpenAI. This note gives explicit finite comparison bounds and shows how the original formulations are related.

1. INTRODUCTION

For an integer $n \geq 2$, let $P(n)$ be the largest prime factor of n , and set $P(1) = 1$. Write ρ for the Dickman–de Bruijn function: the continuous function on $[0, \infty)$ determined by

$$\rho(u) = 1 \quad (0 \leq u \leq 1), \quad u\rho'(u) = -\rho(u-1) \quad (u > 1).$$

The classical smooth-number limit is

$$(1) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{1 \leq n \leq X : P(n) \leq X^c\} = \rho(1/c) \quad (0 < c < 1);$$

see de Bruijn [2, p. 51, (1.1)–(1.2)]. We also use the properties $\rho(u) > 0$ for finite $u \geq 0$ and $\rho(u) \rightarrow 0$ as $u \rightarrow \infty$; see de Bruijn [2, p. 52, (1.8)] and the defining equation.

Throughout, $\mathbb{N} = \{1, 2, \dots\}$. Counting variables are integers, and the cutoff X is real. Natural density means the limit of the unweighted count up to X , divided by X .

Problem 928 in Bloom's collection [1] asks whether the integers satisfying

$$(2) \quad P(n) < n^\alpha, \quad P(n+1) < (n+1)^\beta$$

have a natural density for every fixed $\alpha, \beta \in (0, 1)$. The accompanying discussion asks whether that density equals $\rho(1/\alpha)\rho(1/\beta)$. The cited sources are Erdős's Manitoba and Debrecen papers [3, 4] and the paper of Erdős and Pomerance [5]. We compare their statements in Section 3.

OpenAI's joint Dickman law [6, Theorem 1.1, p. 2] states that, for every fixed $\alpha, \beta \in (0, 1)$,

$$(3) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq n \leq X : P(n) \leq n^\alpha, P(n+1) \leq n^\beta\} = \rho(1/\alpha)\rho(1/\beta).$$

OpenAI states that this theorem resolves the Erdős–Pomerance joint Dickman conjecture. Section 11.3 of the same paper also gives the versions with thresholds X^α, X^β and the corresponding upper tails.

To pass from (3) to (2), one must make the inequalities strict and base the second threshold on the second integer of the pair. Proposition 1 bounds the resulting discrepancy and also treats Erdős's strict formulation. Section 3 identifies the original statements and explains how their inequalities and counting ranges are related.

2. FINITE COMPARISON ESTIMATES

Fix $\alpha, \beta \in (0, 1)$. For $X \geq 2$, define

$$\mathcal{A}_{\alpha, \beta}(X) = \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < (n+1)^\beta\},$$

$$\mathcal{B}_{\alpha, \beta}(X) = \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) \leq n^\alpha, P(n+1) \leq n^\beta\},$$

$$\mathcal{C}_{\alpha, \beta}(X) = \{n \in \mathbb{N} : 2 \leq n \leq X, P(n) < n^\alpha, P(n+1) < n^\beta\}.$$

Here $\mathcal{A}$ is the set in Problem 928, $\mathcal{B}$ is the set counted in (3), and $\mathcal{C}$ is the set in condition (1) of Erdős's Manitoba paper [3, p. 25]. Let $\pi(y) = \#\{p \leq y : p \text{ is prime}\}$, and write $S \triangle T$ for the symmetric difference of two sets.

Proposition 1 (Comparison of conventions). *For every $\alpha, \beta \in (0, 1)$ and every real $X \geq 2$,*

$$(4) \quad \#(\mathcal{A}_{\alpha, \beta}(X) \triangle \mathcal{B}_{\alpha, \beta}(X)) \leq \pi(X^\alpha) + \pi((X+1)^\beta),$$

$$(5) \quad 0 \leq \#\mathcal{B}_{\alpha, \beta}(X) - \#\mathcal{C}_{\alpha, \beta}(X) \leq \pi(X^\alpha) + \pi(X^\beta).$$

Proof. We suppress the subscripts and the argument X throughout the proof. First let $n \in \mathcal{B} \setminus \mathcal{A}$. The inequality $P(n+1) \leq n^\beta < (n+1)^\beta$ shows that the second condition for $\mathcal{A}$ holds, so its first condition must fail: $P(n) = n^\alpha$. Each prime $p = P(n) \leq X^\alpha$ determines at most one such integer, namely $n = p^{1/\alpha}$. Hence $\#(\mathcal{B} \setminus \mathcal{A}) \leq \pi(X^\alpha)$.

Next let $n \in \mathcal{A} \setminus \mathcal{B}$. The first condition for $\mathcal{B}$ is satisfied, so the second must fail:

$$n^\beta < P(n+1) < (n+1)^\beta.$$

The intervals $(m^\beta, (m+1)^\beta)$ for integers $m \geq 2$ are pairwise disjoint, since $\beta > 0$. Thus the map $n \mapsto P(n+1)$ is injective on $\mathcal{A} \setminus \mathcal{B}$. Its image consists of primes below $(X+1)^\beta$, giving $\#(\mathcal{A} \setminus \mathcal{B}) \leq \pi((X+1)^\beta)$. Together, the two bounds prove (4).

Finally, $\mathcal{C} \subseteq \mathcal{B}$, and each $n \in \mathcal{B} \setminus \mathcal{C}$ satisfies $P(n) = n^\alpha$ or $P(n+1) = n^\beta$. As above, the first equality holds for at most $\pi(X^\alpha)$ integers. For the second, $n = P(n+1)^{1/\beta}$ is determined by a prime at most X^β , so at most $\pi(X^\beta)$ integers occur. Adding these bounds proves (5). $\square$

Corollary 2 (Problem 928 and Erdős's strict formulation). *For every fixed $\alpha, \beta \in (0, 1)$,*

$$(6) \quad \lim_{X \rightarrow \infty} \frac{\#\mathcal{A}_{\alpha, \beta}(X)}{X} = \lim_{X \rightarrow \infty} \frac{\#\mathcal{C}_{\alpha, \beta}(X)}{X} = \rho(1/\alpha)\rho(1/\beta).$$

Both the density in Problem 928 and the density $d(\alpha, \beta)$ in Erdős's Manitoba paper [3, p. 25] therefore exist and equal the predicted product.

Proof. For finite sets, $|\#S - \#T| \leq \#(S \triangle T)$. By Proposition 1 and the bound $\pi(y) \leq [y]$, both discrepancies are $O(X^\alpha + X^\beta) = o(X)$. They therefore vanish after division by X , and (3) gives the two limits in (6). $\square$

The comparison above uses no part of the joint Dickman law. For fixed α, β , it measures the error introduced by changing the inequalities and thresholds. The limiting value is supplied by (3); these estimates do not give a convergence rate for that law.

3. COMPARISON WITH THE ORIGINAL SOURCES

We call a threshold *moving* if it depends on the integer being counted, and *fixed* if it remains constant within the counting range. Thus n^α and $(n+1)^\beta$ are moving thresholds, whereas X^α and X^β are fixed thresholds. Both vary with the counting endpoint. Put

$$a(0) = 1, \quad a(t) = 1 - \rho(1/t) \quad (0 < t \leq 1).$$

Table 1 compares OpenAI's theorem with the four source formulations. Its final column gives the normalized limits and the results that establish them. The following subsections explain the inequalities and counting ranges in each source.

TABLE 1. Reference theorem and source formulations. In each row, N counts the integers satisfying the listed conditions. Exponents are fixed: $0 < \alpha, \beta < 1$, $\gamma > 1$, and $0 \leq t, s \leq 1$. Here $c_\gamma = \rho(\gamma)$. Unless indicated otherwise, $2 \leq n \leq X$ and $X \rightarrow \infty$.

Formulation and source	Conditions	Limit and justification
Reference theorem OpenAI [6, Thm. 1.1, p. 2]	$P(n) \leq n^\alpha$ $P(n+1) \leq n^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ Joint Dickman law (3)
Problem 928 Bloom [1]; Erdős's independence statement [3, p. 25]	$P(n) < n^\alpha$ $P(n+1) < (n+1)^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ This note: Proposition 1 and Corollary 2 (set $\mathcal{A}$)
Erdős's strict formulation Manitoba paper, condition (1) [3, p. 25]	$P(n) < n^\alpha$ $P(n+1) < n^\beta$	$\frac{N}{X} \rightarrow \rho(1/\alpha)\rho(1/\beta)$ This note: Proposition 1 and Corollary 2 (set $\mathcal{C}$)
Fixed-threshold pair Erdős (Debrecen), $r = 2$ [4, pp. 272–273]	$P(n+1) \leq k$ $P(n+2) \leq k$ $1 \leq n < k^\gamma, k \rightarrow \infty$	$\frac{N}{k^\gamma} \rightarrow \rho(\gamma)^2 = c_\gamma^2$ OpenAI [6, (11.7), p. 80], with the change of variables in Section 3.2
Fixed upper tails Erdős–Pomerance [5, p. 311]	$P(n) \geq X^t$ $P(n+1) \geq X^s$	$\frac{N}{X} \rightarrow a(t)a(s)$ OpenAI [6, (11.8) and following paragraph, p. 80], including equality and endpoints

3.1. Erdős's Manitoba paper. On page 25 of his Manitoba paper [3], Erdős expects the conditions $P(n) < n^\alpha$ and $P(n+1) < (n+1)^\beta$ to be independent. His next sentence asks whether the integers satisfying condition (1), namely $P(n) < n^\alpha$ and $P(n+1) < n^\beta$, have a density $d(\alpha, \beta)$. The inequalities on Bloom's webpage are those in the independence statement; condition (1) instead uses n^β as its second threshold. Corollary 2 gives the predicted density in both cases.

Positive density also answers Erdős's question about infinitely many solutions for each fixed pair $\alpha, \beta > 0$. For parameters in $(0, 1)$, this follows from (6); for any other positive pair, choose smaller positive parameters inside that interval. As Bloom's webpage [1] notes, infinitude already follows from Schinzel [7, Theorem 14, p. 231], applied to $(2n+1)^2 - 1 = 4n(n+1)$. The theorem gives a constant $K > 0$ and infinitely many integers $n \geq 3$ such that

$$P(n(n+1)) \leq n^{K/\log \log n}.$$

Since $K/\log \log n \rightarrow 0$, these integers satisfy Erdős's strict inequalities for every prescribed positive pair α, β , once n is sufficiently large.

3.2. Erdős's Debrecen paper. Write $\gamma > 1$ for the exponent that Erdős denotes by α in his Debrecen paper [4, pp. 272–273]. He recalls that the number of $m \leq k^\gamma$ satisfying $P(m) \leq k$ is $(c_\gamma + o(1))k^\gamma$, and conjectures the analogous product formula for each fixed integer $r \geq 1$: as k tends to infinity through positive integers,

$$(7) \quad \#\{1 \leq n < k^\gamma : P((n+1) \cdots (n+r)) \leq k\} = (c_\gamma^r + o(1))k^\gamma.$$

The marginal law (1) gives $c_\gamma = \rho(\gamma)$.

The case of two consecutive integers follows from OpenAI's fixed-threshold law [6, (11.7), p. 80]: for fixed $c, d \in (0, 1)$,

$$(8) \quad \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq m \leq X : P(m) \leq X^c, P(m+1) \leq X^d\} = \rho(1/c)\rho(1/d).$$

To apply this law to (7) with $r = 2$, take $X = k^\gamma$, $m = n+1$, and $c = d = 1/\gamma$. Since $P(m(m+1)) = \max\{P(m), P(m+1)\}$, the prime-factor conditions agree with those in (8).

After the shift, the ranges $1 \leq n < X$ and $2 \leq m \leq X$ differ by at most one endpoint term. This difference disappears after normalization, so (8) proves (7) for $r = 2$.

Bloom's webpage [1] also gives a higher-order conjecture with $P(n+i) \leq n^\alpha$ for $0 \leq i < r$. Its case $r = 2$ is precisely (3) with $\beta = \alpha$. The two-variable results used here do not prove the cases $r \geq 3$.

3.3. Erdős and Pomerance. Erdős and Pomerance [5, p. 311] express their independence conjecture in terms of upper tails at fixed thresholds. For $0 \leq t, s \leq 1$, they ask whether

$$(9) \quad b(t, s) = \lim_{X \rightarrow \infty} \frac{1}{X} \#\{2 \leq n \leq X : P(n) \geq X^t, P(n+1) \geq X^s\}$$

exists, is continuous on $[0, 1]^2$, and equals $a(t)a(s)$. Their $a(t)$ denotes the corresponding one-variable upper-tail limit. The marginal law (1) identifies their $a(t)$ with our definition once we include equality and check the endpoints, as below.

For $0 < t, s < 1$, inclusion–exclusion applied to (8) and the two marginal laws gives the limit $a(t)a(s)$ for the strict upper-tail conditions $P(n) > X^t$, $P(n+1) > X^s$. Replacing n by $n+1$ in a marginal count changes its counting range by only $O(1)$ terms, so its limit is unchanged.

Allowing equality leaves the limit unchanged. For fixed $0 < t \leq 1$, equality $P(m) = X^t$ requires X^t to be a prime p . Since $p \mid m$, the number of such integers $m \leq X+1$ is at most $\lfloor (X+1)/p \rfloor = O(X^{1-t} + 1) = o(X)$.

At exponent 0, the upper-tail condition holds automatically for $m \geq 2$. If just one exponent is zero, the marginal law gives the limit for the remaining condition; if both are zero, both conditions are automatic. At exponent 1, the condition $P(m) \geq X$ in the range $2 \leq m \leq X+1$ forces $m \in [X, X+1]$, so at most two integers occur. These observations prove (9) on the boundary as well. The continuity of ρ , together with $\rho(1) = 1$ and $\rho(u) \rightarrow 0$ as $u \rightarrow \infty$, shows that a is continuous on $[0, 1]$. Hence $b(t, s) = a(t)a(s)$ is continuous on the whole square.

OpenAI already gives this argument, including equality and the endpoint cases [6, (11.8) and following paragraph, p. 80]. Here we explain how it yields the limit for the weak upper-tail conditions used by Erdős and Pomerance [5]. Together, the cited joint Dickman law, its fixed-threshold version, and the comparisons above give all four limits in Table 1.

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